

FROM UNDERFITTING TO OVERFITTING: A COMPARATIVE STUDY OF LINEAR AND POLYNOMIAL REGRESSION MODELS

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ABSTRACT

This study examines the impact of model complexity on regression performance by comparing linear and polynomial regression models using World Bank CO₂ emissions per capita data (Bank, 2024). Linear regression is first applied as a baseline to illustrate underfitting, followed by polynomial regression models of increasing degree to investigate improvements in model flexibility. Model performance is evaluated using root mean squared error (RMSE) on both training and testing datasets. The results show that low-degree models fail to capture underlying trends, while high-degree models overfit the data, resulting in poor generalisation. An optimal polynomial degree is identified that balances bias and variance. Furthermore, ridge regularisation is applied to high-degree models, demonstrating improved stability and reduced overfitting (Scikit-learn, n.d.). These findings highlight the importance of selecting appropriate model complexity and regularisation techniques in practical machine learning applications (Géron, 2019 ; Bishop, 2006).

INTRODUCTION

Regression models play a fundamental role in machine learning by enabling the prediction of continuous variables based on input features. Among these, linear regression is widely used due to its simplicity and interpretability; however, it is often limited in capturing complex, non-linear relationships within real-world data (Bishop, 2006). To address this limitation, polynomial regression extends linear models by introducing higher-order terms, thereby increasing model flexibility and enabling more accurate representation of underlying patterns (Géron, 2019).

Despite these advantages, increasing model complexity introduces the risk of overfitting, where a model captures noise rather than meaningful structure, leading to poor generalisation on unseen data. Conversely, overly simple models may underfit

the data, failing to capture essential trends. This trade-off between underfitting and overfitting is a central challenge in machine learning and is closely related to the bias–variance trade-off (Bishop, 2006). Regularisation techniques, such as ridge regression, are commonly employed to mitigate overfitting by constraining model complexity (Scikit-learn, n.d.).

This study investigates these concepts using CO₂ emissions per capita data obtained from the World Bank (Bank, 2024). By implementing linear and polynomial regression models of varying degrees and evaluating their performance using RMSE, this tutorial demonstrates how model complexity affects predictive performance and highlights the importance of selecting an appropriate model for real-world applications.

BACKGROUND THEORY

1. Linear Regression

Linear regression is a fundamental supervised learning method used to model the relationship between an input variable and a continuous output. It assumes a linear relationship, typically expressed as:

$$y = \beta_0 + \beta_1 x$$

where y is the predicted value and x is the input feature (Bishop, 2006). Due to its simplicity and interpretability, it is often used as a baseline model. However, linear regression is limited in its ability to capture non-linear patterns in real-world datasets, such as CO₂ emission trends, which can lead to underfitting and poor predictive performance.

2. Polynomial Regression

Polynomial regression extends linear regression by introducing higher-order terms of the input variable, allowing the model to capture more

complex relationships. The model can be expressed as:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_d x^d$$

where d represents the degree of the polynomial (Géron, 2019). In practical implementation, polynomial features are generated in Python before applying linear regression. Increasing the polynomial degree enhances model flexibility and improves the fit to training data, as observed in the output graphs. However, excessive complexity can cause the model to fit noise rather than meaningful patterns, increasing the risk of overfitting and reducing generalisation performance.

3. Underfitting vs Overfitting

Underfitting and overfitting represent two key challenges in model performance. Underfitting occurs when a model is too simple to capture the underlying structure of the data, resulting in high bias and poor performance on both training and test datasets. In contrast, overfitting occurs when a model becomes overly complex and learns noise from the training data, leading to low training error but high test error. This behaviour is clearly illustrated in the training and testing RMSE curves generated in this study. The balance between these extremes is explained by the bias–variance trade-off, where optimal performance is achieved by selecting an appropriate model complexity (Bishop, 2006).

4. Regularisation (Ridge Regression)

Regularisation techniques are used to control model complexity and reduce overfitting. Ridge regression introduces an L2 penalty term to the loss function, Bishop, 2006, which can be expressed as:

$$\min_{\beta} \sum (y - \hat{y})^2 + \lambda \sum \beta_j^2$$

This penalty discourages large coefficient values, resulting in smoother and more stable model predictions. By reducing the magnitude of coefficients, ridge regression decreases model variance and improves generalisation performance.

In this study, ridge regression is applied to high-degree polynomial models using `Ridge(alpha=10.0)` in Python. As illustrated in Figure 5, the regularised model produces a smoother curve compared to the unregularised model, reducing excessive fluctuations and overfitting.

DATASET DESCRIPTION

The dataset used in this study was obtained from the World Bank's open data repository, which provides global environmental and economic indicators (Bank, 2024). Specifically, the dataset represents annual CO₂ emissions per capita, measured in metric tonnes, for a selected country over a defined time period. CO₂ emissions per capita is a key indicator of environmental impact and reflects changes in industrial activity, energy consumption, and policy decisions over time.

For this analysis, data for a single country was extracted and structured as a time series, where the input feature corresponds to the year and the target variable represents CO₂ emissions per capita. The dataset spans multiple decades, allowing for the observation of long-term trends and variations. As illustrated in the initial output graph, the data exhibits a non-linear pattern with fluctuations, making it suitable for evaluating both simple and more complex regression models.

This dataset is particularly appropriate for regression analysis because it contains continuous numerical values and demonstrates underlying trends that can highlight the effects of underfitting and overfitting when model complexity is varied.

5. Bias–Variance Trade-off

The transition between underfitting and overfitting is explained by the bias–variance trade-off:

$$\mathbb{E}[(y - \hat{y})^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$$

Bias represents error due to simplifying assumptions, while variance reflects sensitivity to training data. As model complexity increases, bias decreases and variance increases. The optimal model balances these two components to minimise

overall error. In this study, the third degree polynomial achieves this balance, as indicated by the lowest test RMSE in Figure 3. Controlling the bias–variance trade-off is essential for achieving reliable generalisation in machine learning (Ghojogh and Crowley, 2019).

METHODOLOGY

1. Data Preparation

The dataset obtained from the World Bank (Bank, 2024) was prepared for regression analysis by selecting a single country’s CO₂ emissions per capita over a defined time period. The input variable was the year, while the target variable was the corresponding CO₂ emissions value. The data was structured as a numerical time series and split into training and testing sets using an approximate 80:20 ratio. This split allows the model to be trained on one subset while evaluating its generalisation performance on unseen data, ensuring a fair comparison between models.

2. Models

Two regression models were implemented using Python and the Scikit-learn library. First, linear regression was used as a baseline model due to its simplicity and interpretability (Bishop, 2006). This model assumes a straight-line relationship between the input and output variables. To capture more complex patterns, polynomial regression models of varying degrees (e.g., 2, 3, 5, 8, and 12) were constructed by transforming the input features into polynomial features before applying linear regression (Géron, 2019). Increasing the polynomial degree allows the model to fit the training data more closely, as observed in the output graphs. However, higher degrees also increase model complexity, which may lead to overfitting and reduced generalisation performance.

3. Evaluation Metric

Model performance was evaluated using Root Mean Squared Error (RMSE), Géron (2019), which is defined as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE measures the average magnitude of prediction errors, with larger errors penalised more heavily. This makes it a suitable metric for regression tasks where accuracy is important. In this study, RMSE was computed using the `mean_squared_error` function in Python and applied to both training and test datasets.

Comparing training and test RMSE allows for clear identification of underfitting and overfitting. As illustrated in Figure 3, the divergence between these errors indicates increasing model complexity and reduced generalisation performance.

RESULTS AND VISUALISATION

1. Raw Data

The initial analysis begins with visualising the CO₂ emissions per capita over time for the selected country. The plot shows a generally non-linear trend, with periods of increase and gradual decline, reflecting changes in industrial activity and environmental policies. This pattern indicates that a simple linear model may not adequately capture the underlying structure of the data.

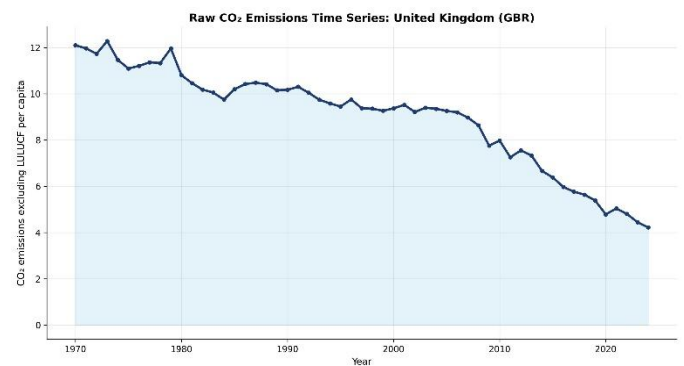


Figure 1: Raw CO₂ Emissions vs Year

2. Linear vs Polynomial

A comparison between linear regression and polynomial regression demonstrates the limitations of simple models. The linear regression model

produces a straight-line fit, which fails to capture the curvature in the data, indicating underfitting. In contrast, a polynomial model with a moderate degree (e.g., degree 3) follows the data more closely and captures the non-linear trend effectively. This highlights the importance of increasing model flexibility to improve performance. However, it also suggests that further increases in complexity must be approached cautiously.

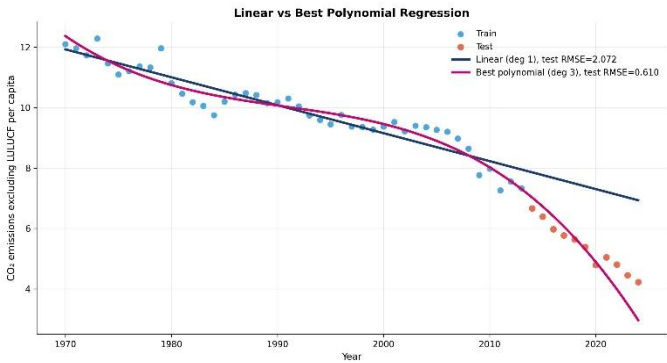


Figure 2: Linear vs Polynomial Regression Fit

3. Model Complexity vs Error

To evaluate the impact of model complexity, the root mean squared error (RMSE) was calculated for both training and testing datasets across different polynomial degrees. As shown in the graph, training error consistently decreases as the model becomes more complex, indicating improved fit to the training data. However, test error initially decreases and then increases beyond a certain degree. This divergence indicates overfitting, where the model begins to capture noise rather than meaningful patterns. The optimal model complexity is identified at approximately degree 3, where test error is minimized, demonstrating the best balance between bias and variance (Bishop, 2006).

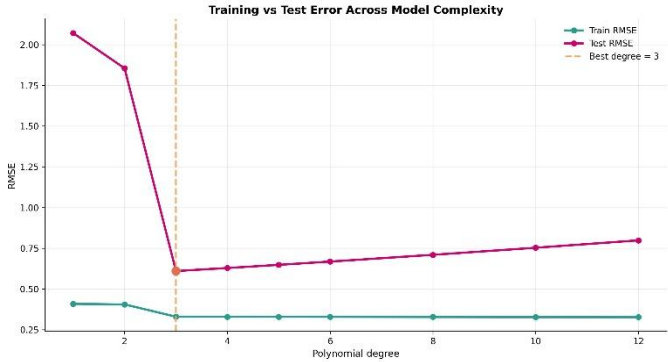


Figure 3: RMSE vs Polynomial Degree

4. Underfitting & Overfitting

A more detailed comparison of model behaviour is illustrated by plotting regression curves for different polynomial degrees. The degree 1 model shows clear underfitting, as it fails to capture the overall trend of the data. The degree 3 model provides a good fit, balancing simplicity and flexibility. However, higher-degree models (degree 8 and degree 12) exhibit increasingly complex curves that closely follow the training data but introduce unrealistic oscillations. These fluctuations are indicative of overfitting, where the model captures noise instead of the true signal. This behaviour aligns with theoretical expectations of the bias-variance trade-off and highlights the importance of selecting an appropriate model complexity (Géron, 2019).

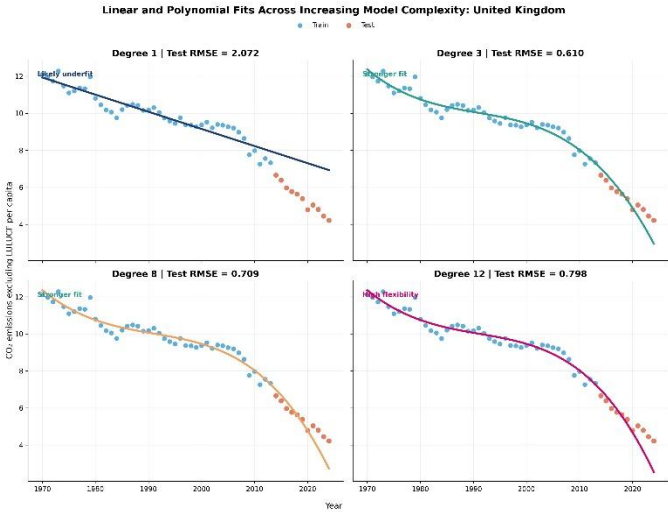


Figure 4: Model Fits for Different Degrees

5. Regularization

To address overfitting in high-degree models, ridge regression was applied as a regularisation technique. The results show that ridge regression produces a smoother curve compared to the unregularised high-degree polynomial model, reducing excessive fluctuations. This indicates that regularisation effectively constrains model coefficients and improves generalisation performance. While the unregularised model fits the training data more closely, the ridge model provides a more realistic representation of the underlying trend. This demonstrates the importance of incorporating regularisation when dealing with complex models (Scikit-learn, n.d.).

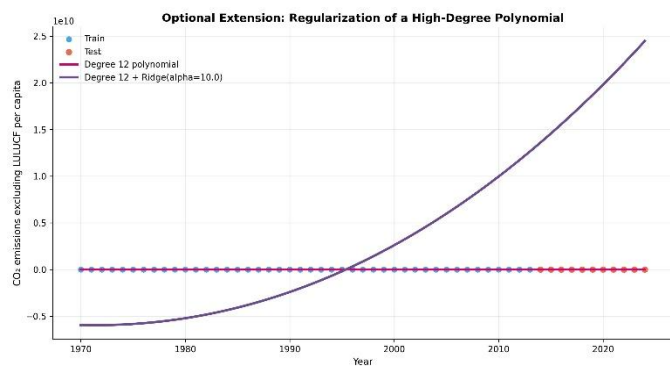


Figure 5: High-Degree Polynomial vs Ridge Regression

DISCUSSION

The results clearly demonstrate the impact of model complexity on regression performance. Underfitting occurs when the model is too simple to capture the underlying structure of the data, as observed with the linear regression model. This behaviour is associated with high bias, where the model makes strong assumptions about the data and fails to represent non-linear patterns present in CO₂ emissions trends (Bishop, 2006).

In contrast, overfitting arises when the model becomes excessively complex, as seen in higher-degree polynomial models. These models achieve very low training error but perform poorly on test data, indicating high variance. This occurs because the model captures noise and fluctuations in the

training data rather than the true underlying relationship (Géron, 2019).

The balance between these two extremes is explained by the bias–variance trade-off, where optimal performance is achieved by selecting an appropriate level of model complexity. In this study, a polynomial degree of approximately three provides the best results, as it minimises test RMSE while maintaining a smooth and realistic fit. This demonstrates how selecting an appropriate model complexity is essential for achieving good generalisation performance in practical applications.

CONCLUSION

This study demonstrated how model complexity influences regression performance by comparing linear and polynomial regression models using CO₂ emissions data (Bank, 2024). The results showed that linear regression underfits the data, while high-degree polynomial models overfit, leading to poor generalisation. An intermediate polynomial degree, particularly degree three, provided the best balance between bias and variance, achieving optimal performance. Furthermore, the application of ridge regularisation improved the stability of high-degree models by reducing overfitting and producing smoother predictions (Scikit-learn, n.d.). These findings highlight the importance of selecting an appropriate model complexity and applying regularisation techniques in practical machine learning tasks.

ACCESSIBILITY CONSIDERATIONS

Several steps were taken to ensure the accessibility of this tutorial. All graphs include clear axis labels, descriptive titles, and legends to enhance readability. Readable font sizes were used throughout the report and visualisations. Colour choices were selected to provide sufficient contrast, ensuring clarity for colour-blind users. Additionally, the figures are designed to be simple and interpretable, allowing readers to easily understand model behaviour and comparisons without requiring advanced technical knowledge.

CODE AVAILABILITY

The code used in this study is available in a public GitHub repository:

<https://github.com/hninhninphyu8895/FROM-UNDERFITTING-TO-OVERFITTING-A-COMPARATIVE-STUDY-OF-LINEAR-AND-POLYNOMIAL-REGRESSION-MODELS-.git>

The repository includes a Jupyter notebook containing all preprocessing, model implementation, and visualisation steps, allowing full reproducibility of the results and figures presented in this tutorial.

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